

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC ASSIGNMENT

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CO5 - AT 2
PSEUDOCODE WRITING TASK

Algorithm Design for Network Performance Analysis

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Declaration

I hereby declare that this assignment is my own original work, prepared under the guidance of my course faculty, and has been submitted in accordance with the academic requirements of the course.

Question 1 - Throughput, Packet Loss and Network Classification

A network simulation records 8000 bits of received data in 4 seconds. Out of 200 packets sent, 180 packets are received. The algorithm must calculate throughput, packet loss percentage, validate the inputs, classify the network, and display a complete performance summary.

Formulae:

- $\text{Throughput} = \text{Total Data Received} / \text{Time} = 8000 / 4 = 2000 \text{ bps}$
- $\text{Packets Lost} = \text{Packets Sent} - \text{Packets Received} = 200 - 180 = 20 \text{ packets}$
- $\text{Packet Loss \%} = (\text{Packets Lost} / \text{Packets Sent}) \times 100 = 10\%$

Pseudocode:

```
BEGIN
    SET dataReceived = 8000
    SET time = 4
    SET packetsSent = 200
    SET packetsReceived = 180

    IF time = 0 THEN
        PRINT "Error: Time cannot be zero"
        STOP
    ENDIF
    IF packetsSent = 0 THEN
        PRINT "Error: Packets sent cannot be zero"
        STOP
    ENDIF

    throughput = dataReceived / time
    packetsLost = packetsSent - packetsReceived
    lossPercent = (packetsLost / packetsSent) * 100

    IF throughput > 1500 AND lossPercent < 10 THEN
        status = "Efficient"
    ELSE
        status = "Inefficient"
    ENDIF

    PRINT "Throughput = ", throughput, " bps"
```

```
PRINT "Packet Loss = ", lossPercent, "%"
PRINT "Network Status = ", status
END
```

Result: The throughput is 2000 bps and packet loss is 10%. Although the throughput is greater than 1500 bps, the loss is not less than 10%; therefore, according to the given condition, the network is classified as Inefficient. Input validation prevents division-by-zero errors and makes the algorithm reliable.

Question 2 - Bandwidth Utilization Monitoring

An organization has several communication links between its branches. The administrator needs to periodically monitor bandwidth utilization and detect overloaded links. The following algorithm stores link utilization values in a list, checks every link continuously, and recommends a backup path whenever utilization exceeds 80%.

Pseudocode:

```
BEGIN
  INPUT numberOfLinks
  CREATE utilization[numberOfLinks]
  CREATE backupStatus[numberOfLinks]

  WHILE monitoring = TRUE DO
    FOR i = 1 TO numberOfLinks DO
      utilization[i] = GET_BANDWIDTH_USAGE(i)
      PRINT "Link", i, "Utilization =", utilization[i], "%"

      IF utilization[i] > 80 THEN
        PRINT "WARNING: Link", i, "is overloaded"
        IF backupStatus[i] = AVAILABLE THEN
          PRINT "Recommendation: Switch traffic to backup link"
          REDIRECT_TRAFFIC(i)
        ELSE
          PRINT "Backup unavailable - reduce non-priority traffic"
        ENDIF
      ELSE
        PRINT "Link", i, "is operating normally"
      ENDIF
    ENDFOR

    WAIT monitoringInterval
  ENDWHILE
END
```

Working and Analysis:

- The loop checks all links at regular intervals, so changing traffic conditions are detected quickly.
- An array/list keeps the latest utilization value of every link and can be used for reports or trend analysis.
- The 80% threshold provides an early warning before a link reaches full capacity and begins dropping or delaying packets.
- Traffic can be shifted to an available backup link, improving load distribution and service continuity.

Conclusion: Continuous bandwidth monitoring helps prevent congestion before it seriously affects users. It improves link utilization, reduces delay and packet loss, and allows the administrator to use backup capacity efficiently while maintaining reliable communication between branches.

Question 3 - Dynamic Best-Path Selection Using Link Quality

A company has six branch offices connected by links whose bandwidth, delay, and utilization change throughout the day. Instead of selecting only the shortest route, the algorithm evaluates overall link quality, detects overload or failure, automatically selects an alternate path, and alerts the administrator.

Link Quality Model:

A higher bandwidth should increase the score, while higher delay and utilization should reduce it. One simple normalized model is: $\text{Quality Score} = (\text{Bandwidth} \times 0.5) - (\text{Delay} \times 0.3) - (\text{Utilization} \times 0.2)$. The exact weights can be adjusted according to organizational priorities.

Pseudocode:

```
BEGIN
  SET branches = 6
  CREATE bandwidth[], delay[], utilization[], score[]

  WHILE networkActive = TRUE DO
    FOR each available path p DO
      bandwidth[p] = GET_BANDWIDTH(p)
      delay[p] = GET_DELAY(p)
      utilization[p] = GET_UTILIZATION(p)

      IF LINK_FAILED(p) = TRUE OR utilization[p] >= 90 THEN
        score[p] = -1
        PRINT "ALERT: Path", p, "failed or overloaded"
      ELSE
        score[p] = (0.5 * bandwidth[p])
                  - (0.3 * delay[p])
                  - (0.2 * utilization[p])
      ENDIF
    ENDFOR

    bestPath = PATH_WITH_HIGHEST_SCORE(score)
    IF bestPath EXISTS THEN
      ROUTE_TRAFFIC(bestPath)
      PRINT "Selected Path =", bestPath
    ELSE
      PRINT "CRITICAL ALERT: No usable path available"
```

ENDIF

WAIT monitoringInterval

ENDWHILE

END

Explanation: The algorithm repeatedly gathers current measurements instead of relying on fixed routing information. A good path receives a higher score because it combines sufficient bandwidth with low delay and moderate utilization. Failed or severely overloaded links are removed from selection, so an alternate path is chosen automatically.

Conclusion: This adaptive approach improves routing efficiency, network reliability, and fault tolerance. It responds to changing conditions in real time, avoids congested links, maintains communication with branch offices, and gives the administrator immediate alerts when abnormal conditions occur.